

Towards Utilizing Touch-sensitive Physical Keyboards for Text Entry in Virtual Reality

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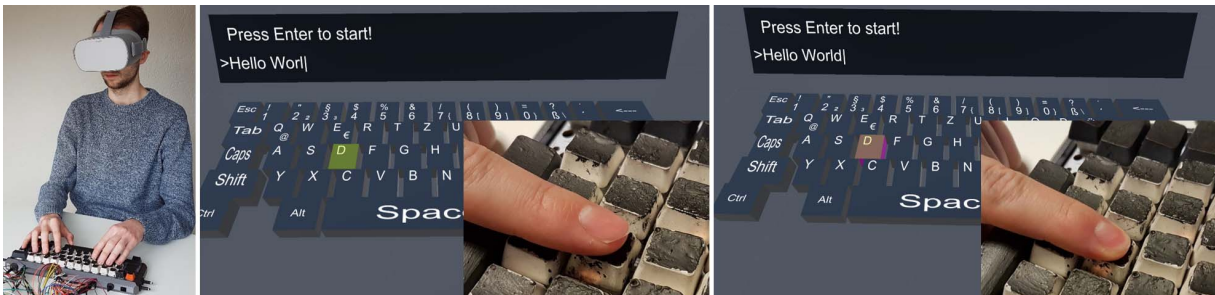


Figure 1: View on the virtual text entry environment and the touch-sensitive physical keyboard. Left: A user typing on the physical keyboard. Middle: Touch state of a key. The respective virtual key is highlighted in green. Right: Pressed state of a key. The respective virtual key is highlighted in purple.

ABSTRACT

Text entry is a challenge for Virtual Reality (VR) applications. In the context of immersive VR Head-Mounted Displays, text entry has been investigated for standard physical keyboards as well as for various hand representations. Specifically, prior work has indicated that minimalistic fingertip visualizations are an efficient hand representation. However, they typically require external tracking systems.

Touch-sensitive physical keyboards allow for on-surface interaction, with sensing integrated into the keyboard itself. However, they have not been thoroughly investigated within VR.

Our work brings together the domains of VR text entry and touch-sensitive physical keyboards. Specifically, we propose to utilize touch-sensitive physical keyboards for text entry as an alternative sensing mechanism for tracking user's fingertips and study its performance in a preliminary user study.

Index Terms: H.5.2: [User Interfaces - Input devices and strategies.]

1 INTRODUCTION

Text entry in Virtual Reality (VR) is an important feature for many tasks, such as note taking, messaging and annotation. The challenge of realizing efficient and usable text entry systems is deemed crucial for realizing future virtual office environments [6]. Existing consumer-grade immersive head-mounted display (HMD) VR systems, often rely on indirect control of a virtual pointer using hand-held controllers or head or gaze direction. However, these methods

are limited in performance and consequently mostly used to enter short texts, such as passwords or names. Further, they require some degree of training due to their unfamiliarity to some users, compared to standard desktop and touchscreen keyboards that users are already familiar and proficient with.

Hence, recently, several groups investigated the feasibility of utilizing standard keyboards for text entry in immersive HMD-based VR focusing on different hand representations, including no hand representations [29], minimalistic fingertip rendering [7, 8], video see-through [2, 19] to full inverse kinematic hand models [?, 17]. In 2017, Logitech made a toolkit available which utilizes a blended camera view [2, 28]. However, apart from using no hand model (which results in unacceptable high error rates without auto-decoders [29]), all proposed techniques rely on the availability of external tracking systems integrated into the environment (e.g., using infrared-based outside-in tracking systems) or in the HMD (e.g., using cameras). As cameras or external hand tracking systems are not widely integrated into the majority of HMDs, this can potentially hinder the widespread use of text entry on physical keyboards.

Our work extends these previous works by investigating if efficient text entry in VR can be achieved through an alternative sensing modality. Specifically, we look at touch-sensitive keyboards through capacitive sensing, which allow to visualize a fingertip on a physical key before a character has been entered. In fact, there is a large body of work on touch-sensitive keyboards using capacitive sensing that has been investigated outside of VR (e.g., [1, 10, 22, 25, 26]). It has been demonstrated that touch-sensitive keyboards have the potential to be commercialized (e.g., [4]).

In this paper, we focus on utilizing touch-sensitive physical keyboards for efficient text entry in VR as a tool for enabling portable virtual work environments [6]. We envision that future office workers only need access to a laptop with integrated touch-sensing keyboard and trackpad as well as a untethered VR Head-Mounted Display (HMD) for working productively on the go. Specifically, we do not assume any tracking capability of the mobile HMD beyond 3 degrees of freedom rotation sensing. As a first step towards realizing this vision it is crucial to understand if future mass-manufactured touch-sensitive physical keyboards lend themselves for efficient text entry in VR.

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We are focusing on physical keyboards as research has shown that typing on those physical keyboards substantially outperforms text entry on multi-touch tablets, which could be considered an alternative text entry method [8].

In this work, we built a physical keyboard with capacitive sensing, which allows to sense and visualize the user's finger tip before a key has been pressed. This could be beneficial for virtual office work on the go.

2 RELATED WORK

Our work draws on the domain of text entry in VR, specifically typing on keyboards, as well on the domain of touch-sensitive physical keyboards.

2.1 Text Entry in VR

Text entry has been extensively researched (see [18, 31] for surveys and overviews). However, relatively few text entry methods have been proposed for VR.

Prior work has investigated a variety of wearable gloves, where touching between the hand fingers may represent different characters or words, e.g., [3, 13, 30]. Such devices enable a mobile, eyes-free text entry. However, most of them require a considerable learning effort, and may limit the user ability to use other input devices while interacting in VR, such as game controllers.

The mainstream mobile phone touchscreen keyboard might be a good text entry option [5]. Recent work has investigated the feasibility of using multi-touch screens for text entry in VR, e.g., using a smartphone [16], a face-mounted display [9] or a multi-touch tablet [8]. Specifically, Grubert et al. [8] indicated a relatively low text entry performance using multi-touch tablets, Guggenheimer et al. [9] and Kim et al. [16] did not report standard text entry metrics.

While we do not investigate this small type of touchscreen keyboard in this paper, we do believe it has potential.

Recent research has investigated the feasibility of typing on a physical full-sized keyboard in VR.

An obvious problem is the lack of visual feedback. Without visual feedback users' typing performance degraded substantially. However, by blending video of the user's hands into VR the adverse performance differential significantly reduced [19].

Fundamentally, there are three solution strategies for supporting keyboards in VR. First, by providing complete visual feedback by blending the user's hands into VR. Second, by decoding (auto-correcting) the user's typing to compensate for noise induced by the lack of feedback. Third, by investigating hybrid approaches, such as minimal visual feedback, which may or may not require a decoder to compensate for any noise induced by the method.

Walker et al. [29] investigated typing on a physical keyboard with no visual feedback. They found that the character error rate (CER) was unacceptably high but could be reduced to an average 3.5% CER using an auto-correcting decoder.

McGill et al. [19] investigated typing on a physical keyboard in Augmented Virtuality [20] with camera view on a physical keyboard.

The closest work to ours is from Grubert et al. [7, 8]. In 2018, they investigated the performance of physical and touch keyboards and physical/virtual co-location for VR text entry. They proposed to use minimalistic fingertip rendering as hand representation and indicated that this representation is as efficient as a video-see through of the user's physical hands [7]. Subsequently, similar studies have investigated further hand representations, such as semi-transparent hand models [17]. Besides, optical outside-in tracking systems with sub-millimeter accuracy such as Optitrack Prime series (e.g., in [7, 8, 17]), also commodity tracking devices such as the Leap Motion have been utilized [12] for hand tracking on physical keyboards.

Our work extends these previous works by investigating if efficient text entry in VR can be achieved through an alternative sensing modality. Specifically, we look at touch-sensitive keyboards through

capacitive sensing, which allow to visualize a fingertip on a physical key before a character has been entered.

2.2 Touch-sensitive physical keyboards

There have been several approaches in extending the basic input capabilities of physical keyboard beyond individual button presses. Specifically, embedding capacitive sensing into keyboards has been studied by various researchers. It lends itself to detect finger events on and slightly above keys and can be integrated into mass-manufacturing processes. Habib et al. [10] and Tung et al. [26] proposed to use capacitive sensing embedded into a full physical keyboard to allow touchpad operation on the keyboard surface. Shi et al. developed microgestures on capacitive sensing keys [24, 25].

Extending the idea of LCD-programmable keyboards [14], Block et al. extended the output capabilities of touch-sensitive, capacitive sensing keyboard by using a top-mounted projector [1].

Our work utilizes capacitive sensing for detecting resting fingers on individual keys, which notifies a user of the selected key before a character is entered through a key press.

3 TOUCH-SENSITIVE KEYBOARD FOR TYPING IN VR

We detect key touches on individual keys through capacitive sensing. A diagram portraying the basic principle of the keyboard is shown in Figure 2. Each key is coated with conductive electric paint that is connected to a specific digital pin on an external microcontroller via a copper wire on the underside of the keyboard. The conductive surface of each key together with the environment around the surface functions as a capacitor whose physical properties change when the environment around the key changes, for example when the user touches the key with a finger. In software, when the pin that is connected to a key is set to high via the internal pull up mechanism of the microcontroller, the time until the signal on the pin physically reaches high differs depending on the physical properties of the capacitive system between the key and its environment. Once at application startup, we measure a baseline time delay for the signal to reach high for each key when the capacitive system consists of the key and an empty air environment, by setting the pin to high and then continuously reading its value until it returns high as well. At runtime, we then continuously measure the time delay for the signal to reach a high value for each key, which changes when the user touches a key as an effect of the physical properties of the capacitor changing. If the relative difference of this time delay to the measured baseline delay reaches an empirically determined threshold of 50%, the key is assumed to be touched by the user.

The microcontroller is connected to an external PC via a standard USB serial connection that is used for communication between the client application running the experiment software and the microcontroller that detects key touches. Each physical key on the keyboard is connected to a separate pin on the microcontroller, and when a key touch is recognized on each respective pin, a serial message is sent to the client application containing information about which key was touched.

The keyboard uses two Arduino Mega 2560 Rev3 microcontrollers, due to the number of separate keys surpassing the number of pins available on a single microcontroller. The keyboard itself consists of a GK64 hot swappable PCB using a custom 60% keyboard layout, using Kailh BOX Pale Blue as mechanical switches for each key, Tai-Hao Carbon Keycap Set ISO-DE keycaps and a 5 KBDfans 60% case as a casing for the entire keyboard. Overall, the keyboard has 62 keys, with each alphanumeric key having dimensions of 12 x 14 mm (width x height).

In VR, the respective key of a virtual keyboard is highlighted as the user touches the key (see Figure 1, middle and right).

4 PRELIMINARY USER STUDY

We investigated the performance when writing with the touch-sensitive keyboard with a fingertip visualization, that was found

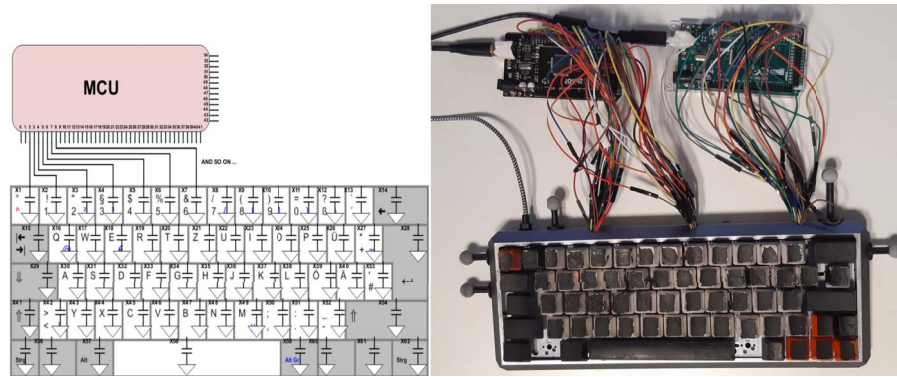


Figure 2: Left: Circuit diagram showing the function principle of the keyboard's capacitive sensing scheme. Each of the microcontroller unit's (MCU's) binary pins is connected to one of the capacitors denoted as x_1 through x_{62} , where each capacitor consists of the key and its physical environment. The layout only shows the logical structure and does not represent the actual physical layout of the keyboard. Right: Overhead view of the actual physical keyboard as used in the experiments.

the be an efficient representation in VR text entry [7] in a controlled experiment. We used a within-subjects design with a single independent variable—VISUALIZATION—with two levels: highlighting of a touched key (CAPACITIVE TOUCH), which made use of the capacitive sensing capability of the keyboard and fingertip visualization through spheres (FINGERTRACKING), which made use of an external tracking system. FINGERTRACKING was employed as baseline. We recruited 6 participants (2 female, 4 male, mean age 27.7 years, $sd = 3.7$) from a university campus. All participants were familiar with QWERTZ desktop keyboard typing and QWERTZ touchscreen keyboard typing.

4.1 Procedure and Task

After welcoming, participants were asked to fill out a demographic questionnaire. Thereafter, we conducted a calibration phase, in which participants entered text for 5 minutes for profiling reasons and in which retro-reflective markers were attached to the fingernails.

The order of the conditions was balanced across participants. In either condition, participants were shown a series of stimulus sentences. For an individual stimulus sentence, participants were asked to type it as quickly and as accurately as possible. Participants typed stimulus sentences for 15 minutes in each condition. Participants were allowed to use the backspace key to correct errors.

The conditions were separated by a 5-minute break, in which participants filled out a the SSQ simulator questionnaire [15], the NASA TLX questionnaire [11], the IPQ [21] spatial presence questionnaire and the Flow-Short-Scale [23]. After the final condition, participants were asked to fill out a self-assessment questionnaire, which was followed by a semi-structured interview. The experiment was carried out in a 90-minute session.

Stimulus sentences were drawn from the mobile email phrase set [27], which provides a set of text entry stimulus sentences that have been verified to be both externally valid and easy to memorize for participants. Participants were shown stimulus phrases randomly drawn from the set.

An OptiTrack Prime 13 outside-in tracking system was used for spatial tracking of finger tips, the HMD, an external monitor and the keyboard. The monitor and keyboard were tracked for logging reasons. An Oculus Rift CV1 was used as HMD. The physical keyboard described in section 2.2 was used. The virtual environment was showing the virtual keyboard and a monitor resting on a desk. In line with previous work [7, 8], the phrases and the entered text were visualized both on the monitor and directly above the keyboard.

4.2 Results

The performance results are summarized in Figure 3.

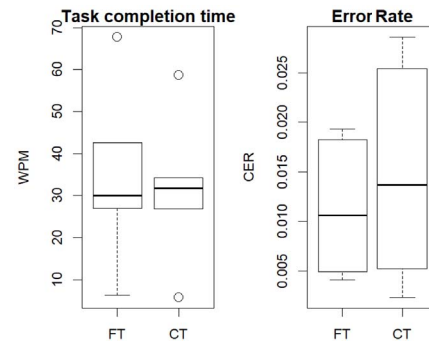


Figure 3: Left: Task completion time in word-per-minute. Right: Character Error rate. FT: FINGERTRACKING, CT: CAPACITIVE TOUCH

Statistical significance tests for entry rate and error rate were carried out using paired, two-tailed t-tests with a significance level of $\alpha = 0.05$. The analysis was checked for appropriateness against the dataset, log transformed data was used for error rate. Subjective feedback was analyzed with non-parametric Wilcoxon signed-rank tests.

Entry rate was measured in words-per-minute (wpm), with a word defined as five consecutive characters, including spaces. The entry rate was 33.2 wpm ($sd = 20.3$) for FINGERTRACKING and 31.5 wpm ($sd = 17.0$) for CAPACITIVE TOUCH. The difference in entry rate was not significant ($t(5) = 1.18$, $p = 0.293$).

Error rate was measured as character error rate (CER). CER is the minimum number of character-level insertion, deletion and substitution operations required to transform the response text into the stimulus text, divided by the number of characters in the stimulus text. The character error rate (CER) was 1.1% ($sd = 0.07$) for FINGERTRACKING and 1.5% ($sd = 0.1$) for CAPACITIVE TOUCH. The difference in error rate was not significant ($t(5) = -0.31$, $p = 0.76$). No significant differences could be indicated for SSQ, NASA TLX, IPQ or the Flow-Short-Scale.

5 DISCUSSION AND CONCLUSION

Touch-sensitive physical keyboards allow for on-surface interaction, with sensing integrated into the keyboard itself.

Our study indicates that writing on a touch-sensitive keyboard has the potential to be as efficient as typing using fingertip visualization, and, hence, external tracking or camera systems might not be needed to support text entry in VR, which could be beneficial for virtual office work on the go.

However, the small number of participants do not allow to easily generalize those findings to the population. Also, it is unclear how typing proficiency could influence typing performance on a touch-sensitive keyboard. Please note, that our keyboard was not using the standard 104-key layout due to lack of availability of compatible PCBs. Typing on this smaller keyboard could result in absolute performance differences compared to a larger keyboard, but for both conditions. Hence, the relative performance differences (or the lack thereof) can be seen as transferable to other keyboards.

Our work is a first step in making touch-sensitive keyboards available for VR text entry. If touch-sensitive keyboards will become commercially widely available they could become an efficient tool for VR text entry. In future work, we want to thoroughly investigate the performance envelope of touch-sensitive physical keyboards.

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